Overview of the BIOMASS R package

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GEO-TREES Panama Workshop

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Brief introduction

For what ?

Estimate Aboveground Biomass and Its Uncertainty in Tropical Forests

$\rightarrow$ contains functions for estimating above-ground biomass/carbon and its uncertainty in tropical forests. These functions allow to:

1. correct taxonomy (typos & synonyms),
2. estimate wood density and its uncertainty,
3. build height-diameter models,
4. manage tree and plot coordinates,
5. estimate above-ground biomass/carbon at stand level with associated uncertainty,
6. predict and map biomass from LiDAR data with associated uncertainty.

CRAN R package

Current CRAN version: 2.2.7

BIOMASS: Estimating Aboveground Biomass and Its Uncertainty in Tropical Forests

Contains functions for estimating above-ground biomass/carbon and its uncertainty in tropical forests. These functions allow to (1) retrieve and correct taxonomy, (2) estimate wood density and its uncertainty, (3) build height-diameter models, (4) manage tree and plot coordinates, (5) estimate above-ground biomass/carbon at stand level with associated uncertainty. To cite 'BIOMASS', please use citation('BIOMASS'). For more information, see Réjou-Méchain et al. (2017) <doi:10.1111/2041-210X.12753>.

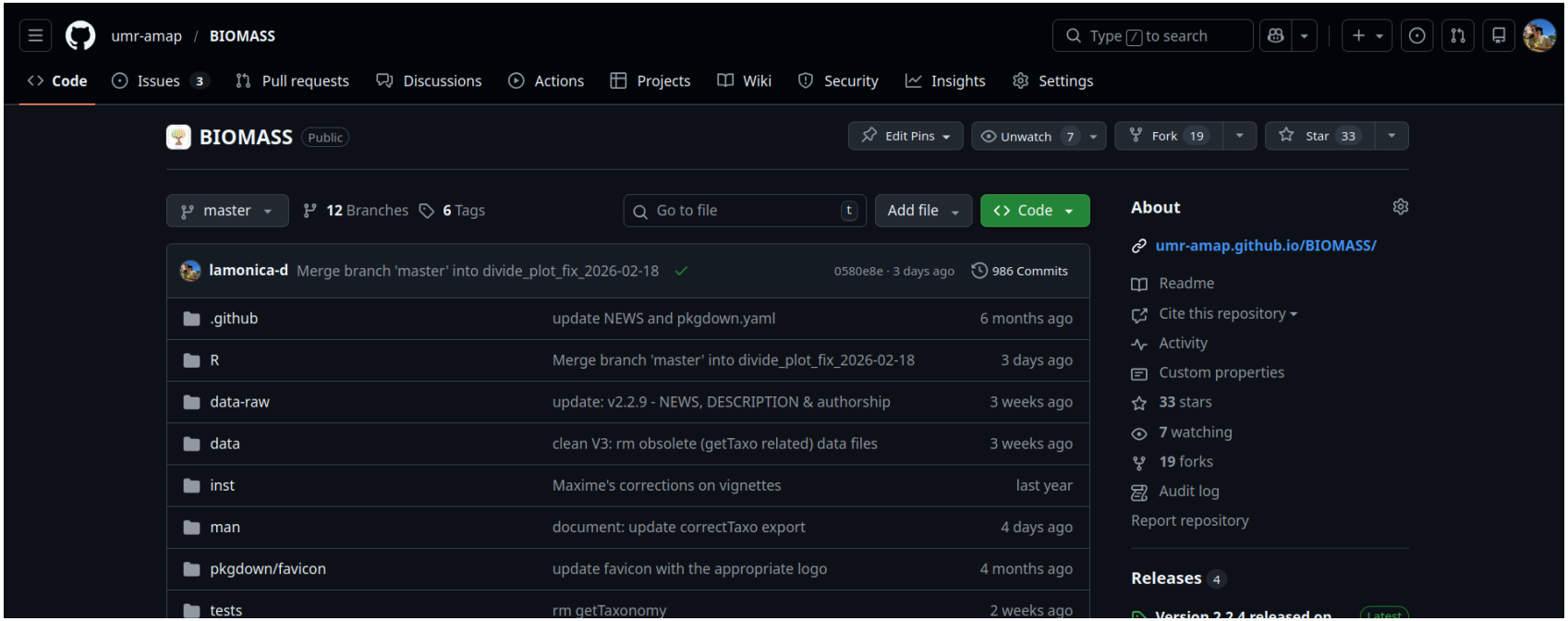
Version:	2.2.7
Depends:	R (≥ 3.6)
Imports:	minpack.lm, jsonlite, methods, proj4, graphics, stats, utils, data.table (≥ 1.9.8), rappdirs, sf, terra, ggplot2
Suggests:	knitr, rmarkdown, prettydoc, testthat, vdiffr, curl, geodata, httr2, pkgdown, dplyr, brms, BH, RcppEigen
Published:	2026-03-11
DOI:	10.32614/CRAN.package.BIOMASS
Author:	Dominique Lamonica [aut, cre], Maxime Réjou-Méchain [aut, dtc], Arthur Bailly [aut], Guillaume Cornu [aut], John Godlee [ctb], Fabian Fischer [ctb], Jerome Chave [ctb], Arthur Pere [aut], Ariane Tanguy [aut], Camille Piponiot [aut], Bruno Hérault [aut], Philippe Verley [ctb], Ted Feldpausch [dtc]

Archives

BIOMASS_1.0.tar.gz	2016-09-21 19:27	749K
BIOMASS_1.1.tar.gz	2017-01-05 11:55	1.3M
BIOMASS_1.2.tar.gz	2018-02-21 09:33	1.3M
BIOMASS_2.0.tar.gz	2019-01-14 11:30	807K
BIOMASS_2.1.1.tar.gz	2019-05-03 17:20	916K

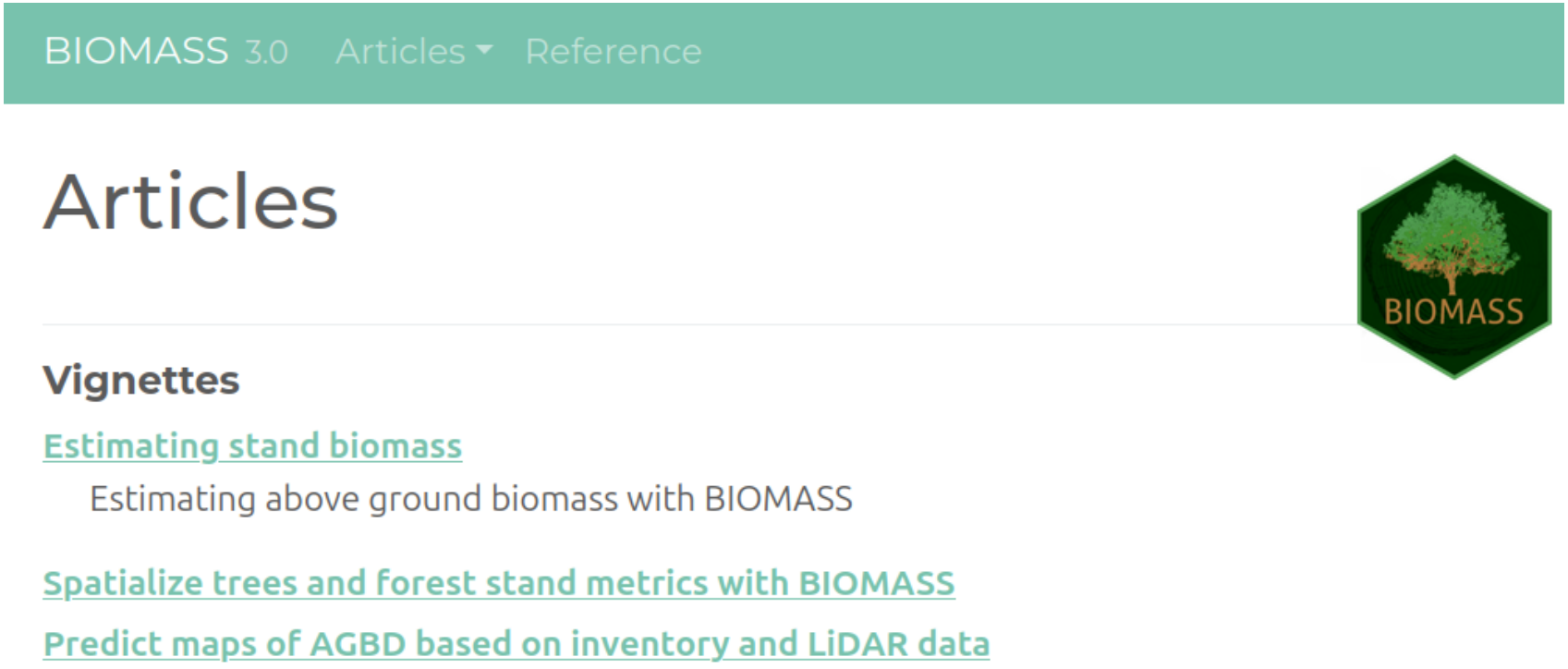
Almost 10 years of existence!

GitHub public repository

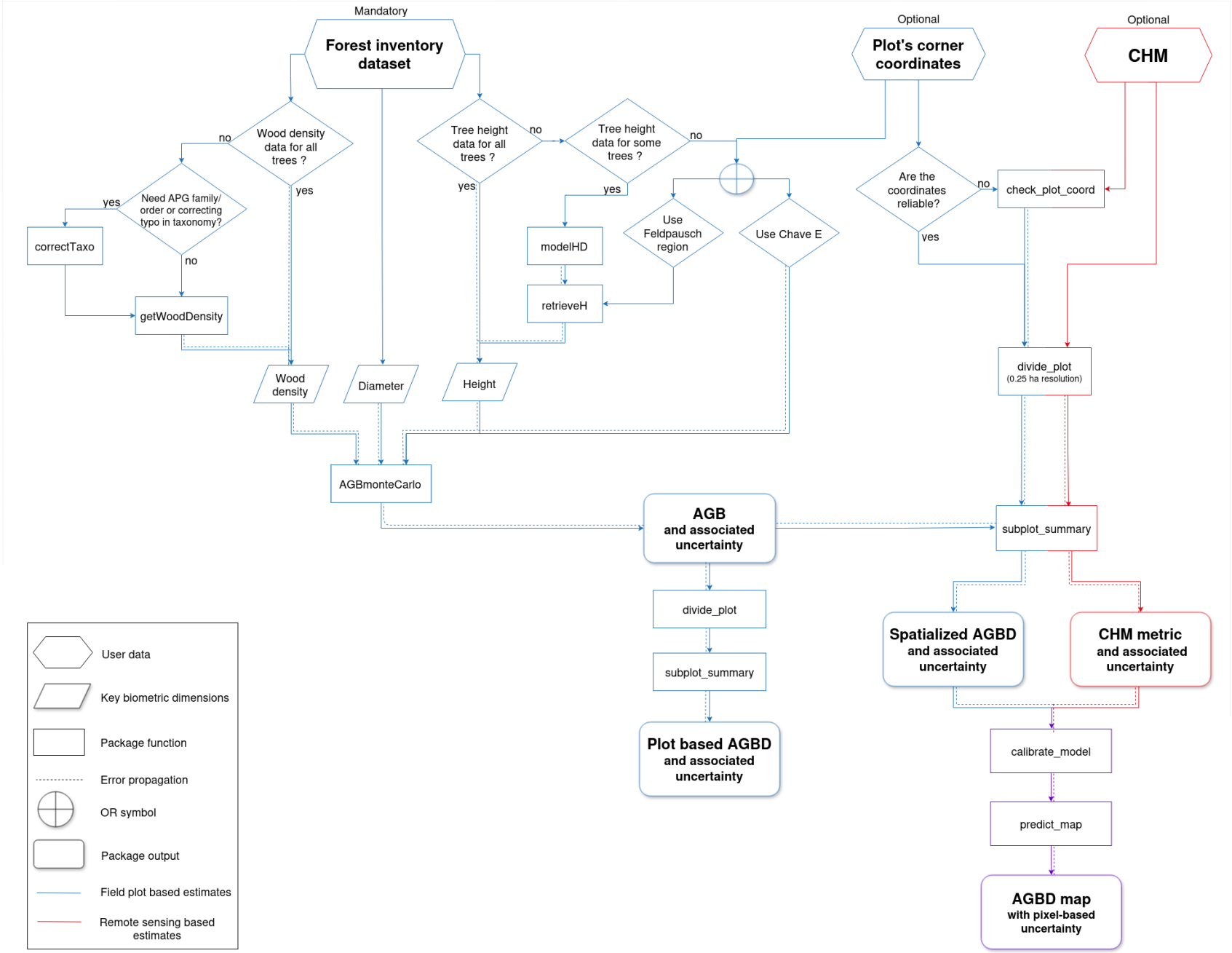


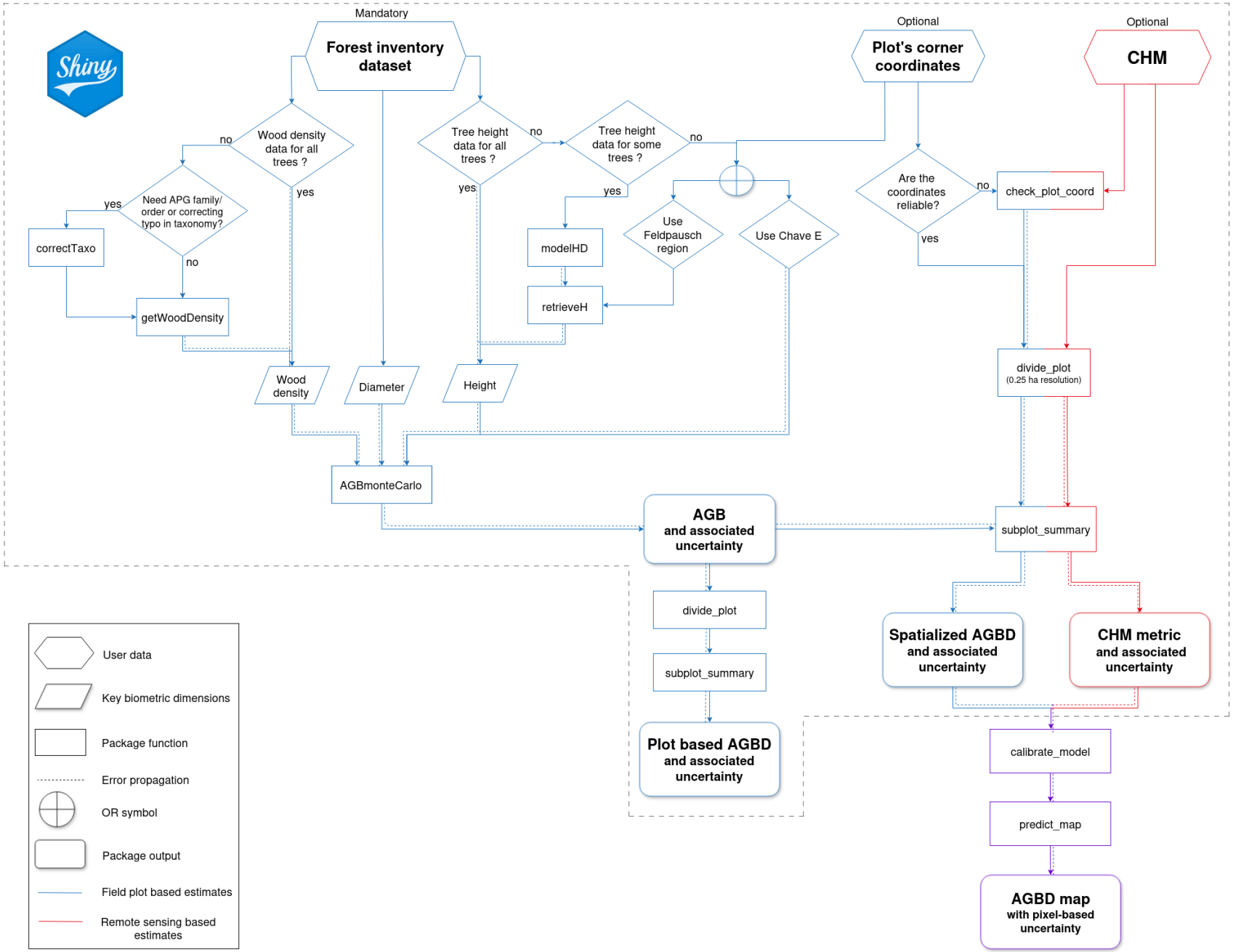
- Entire code accessible, issues open
- Developing version: 3.0
- Continuous integration

Vignettes



- Tutorial for the entire pipeline divided in three main articles, applied on reduced Nouragues data (French Guiana)


Mandatory
Forest inventory dataset
Optional
Plot's corner coordinates
Optional
CHM
Wood density data for all trees ?
no
yes
Need APG family/ order or correcting typo in taxonomy?
yes
no
correctTaxo
getWoodDensity
Tree height data for all trees ?
no
yes
Tree height data for some trees ?
no
yes
modelHD
retrieveH
Use Feldpausch region
Use Chave E
Are the coordinates reliable?
no
yes
check_plot_coord
Wood density
Diameter
Height
divide_plot
(0.25 ha resolution)
AGBmonteCarlo
AGB
and associated uncertainty
subplot_summary
divide_plot
subplot_summary
Plot based AGBD
and associated uncertainty
Spatialized AGBD
and associated uncertainty
CHM metric
and associated uncertainty
calibrate_model
predict_map
AGBD map
with pixel-based uncertainty
User data
Key biometric dimensions
Package function
Error propagation
OR symbol
Package output
Field plot based estimates
Remote sensing based estimates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Shiny app - BIOMASSapp

- Estimate AGBD per subplot with uncertainties, and spatialize stand metrics
- No map prediction: target users & technical limitations

$\rightarrow$ **Why such a tool ?**

- Need for a graphical, user-friendly interface for non coding and/or fieldwork people...
- ... actually even R people are using it! It helps for data checks, or visualization for example.
- Issues or feedback to be reported on the GitHub repository

BIOMASSapp demonstration

Predict AGBD maps using inventory and LiDAR data

AGBD-LiDAR modelisation: proposed statistical framework

- geostatistical model with SPV-I/C (SPatially Varying Intercept/Coefficients) to integrate spatial correlation:
- $y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$
  with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$
- references

Remote Sensing of Environment 169 (2015) 113–127

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

LiDAR based prediction of forest biomass using hierarchical models with spatially varying coefficients

Chad Babcock [a], Andrew O. Finley [b,*], John B. Bradford [c], Randall Kolka [d], Richard Birdsey [e], Michael G. Ryan [f]

Remote Sensing of Environment 318 (2025) 114557

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

A geostatistical approach to enhancing national forest biomass assessments with Earth Observation to aid climate policy needs

Neha Hunka [a,*], Paul May [b], Chad Babcock [c], José Armando Alanís de la Rosa [d], Maria de los Ángeles Soriano-Luna [d], Rafael Mayorga Saucedo [d], John Armston [a], Maurizio Santoro [e], Daniela Requena Suarez [f], Martin Herold [f], Natalia Málaga [f], Sean P. Healey [g], Robert E. Kennedy [h], Andrew T. Hudak [i], Laura Duncanson [a]

Implementation in BIOMASS

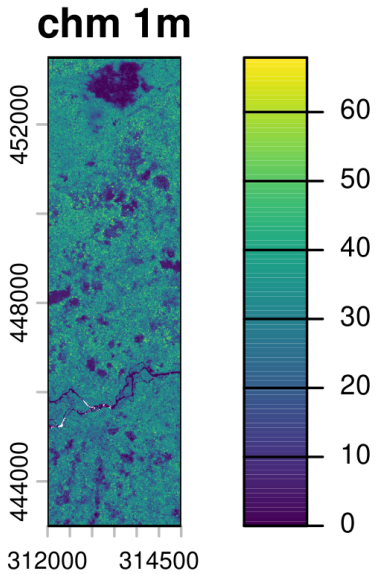
- inference and predictions done using brms R package
- two functions: 1st calibrates the model with user plot and LiDAR data, 2nd predicts AGBD and its associated uncertainties over the full raster footprint provided by the user
- full continuity of the pipeline: the inputs of those functions are the outputs from the previous steps
- propagate all AGBD uncertainties from the previous steps, within Bayesian framework

Example - Nouragues: input data

Forest Inventory Data Preview

how 10 entries Search:

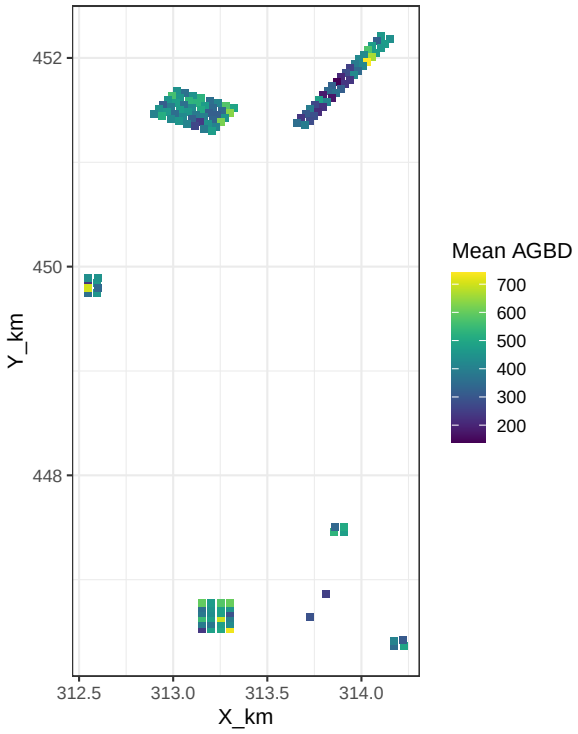
	Site	Plot	Xfield	Yfield	Family	Genus	Species	D
1	Petit_Plateau	201	0	31.5	Burseraceae	Protium	surinamense	11
2	Petit_Plateau	201	0.100000001490116	75.1999969482422	Anacardiaceae	Tapirira	guianensis	74.4
3	Petit_Plateau	201	0.200000002980232	27.6000003814697	Lecythidaceae	Indet.Lecythidaceae	Indet.	25.4
4	Petit_Plateau	201	-4	67.5	Euphorbiaceae	Conceveiba	guyanensis	10
5	Petit_Plateau	201	0.300000011920929	39.9000015258789	Burseraceae	Protium	altissimum	18.9
6	Petit_Plateau	201	-3.5	41.5	Euphorbiaceae	Mabea	speciosa	10
7	Petit_Plateau	201	0.5	78.5	Fabaceae	Paloue	princeps	15.9
8	Petit_Plateau	201	0.899999976158142	55	Rubiaceae	Amaioua	guianensis	10.1

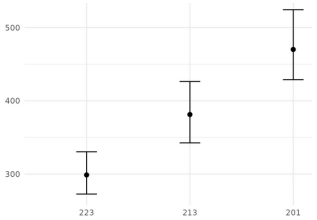


- Tree inventory & (including coordinates)
- LiDAR metric landscape level (e.g. CHM)

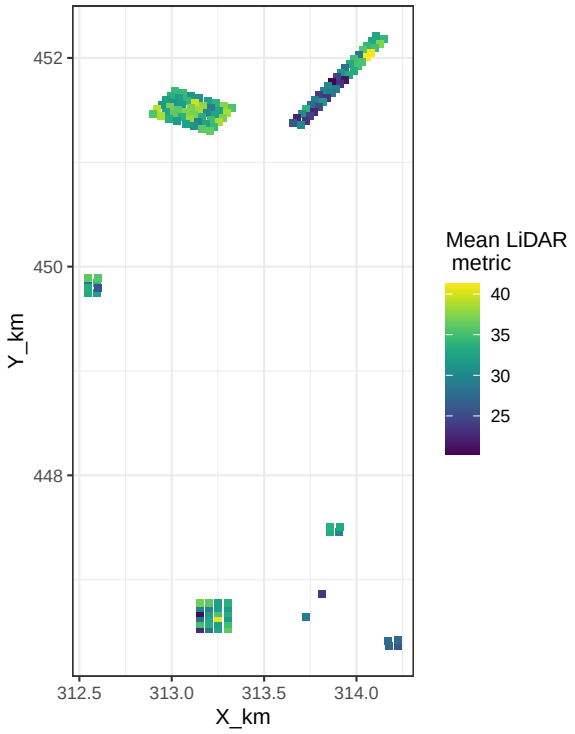
→ BIOMASS pipeline to run...

… to get predicted variable y & predictive variable x





$\sim$

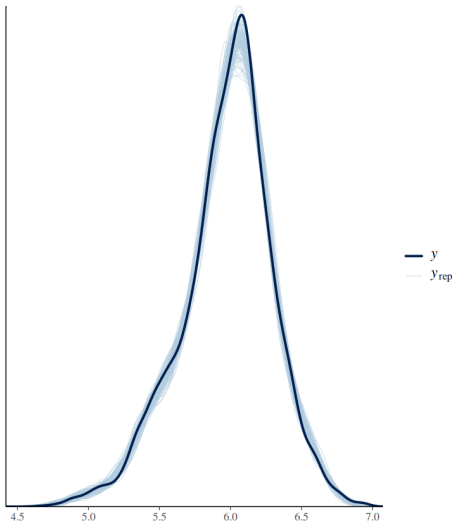


- y AGBD 0.25ha for each (sub)plot with uncertainties
- x mean CHM 0.25ha for each (sub)plot
- distances between plots

CHM-AGBD model calibration

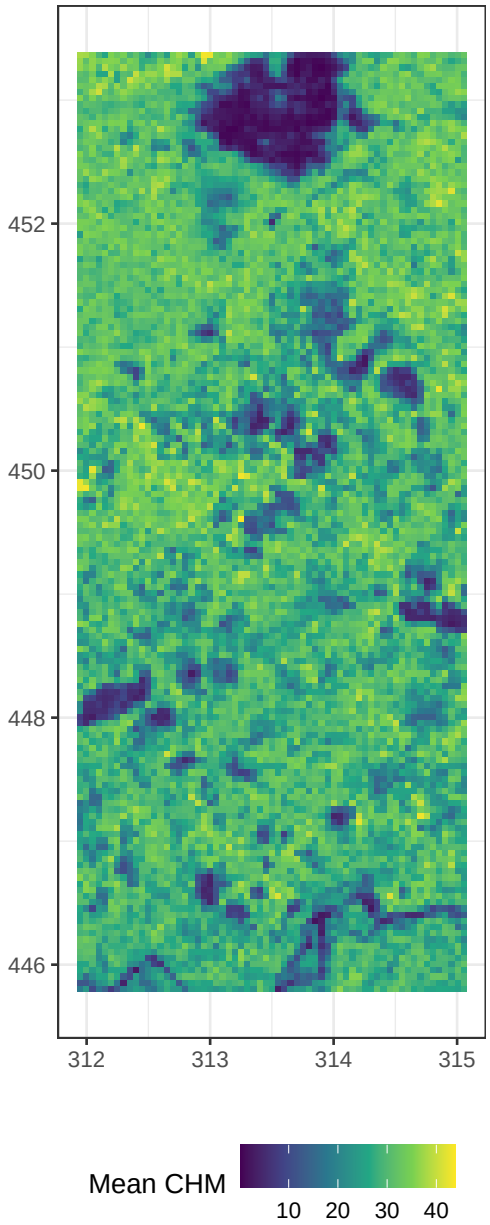
The function returns a *brms.fit* object

```
 Family: gaussian
  Links: mu = identity
Formula: log_AGBD ~ 0 + betatilde * mean_logCHM
         betatilde ~ 1 + gp(X_km, Y_km, gr = T, scale = T, cov = "matern32")
   Data: df_inf (Number of observations: 3900)
  Draws: 5 chains, each with iter = 2300; warmup = 300; thin = 20;
         total post-warmup draws = 500

Gaussian Process Hyperparameters:
                              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sdgp(betatilde_gpX_kmY_km)       0.07      0.00     0.06     0.08 1.00      349      426
lscale(betatilde_gpX_kmY_km)     0.01      0.00     0.00     0.01 1.01      302      317

Regression Coefficients:
                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
betatilde_Intercept     1.73      0.01     1.71     1.75 1.01      203      333

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.13      0.00     0.12     0.13 1.00      580      531

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```



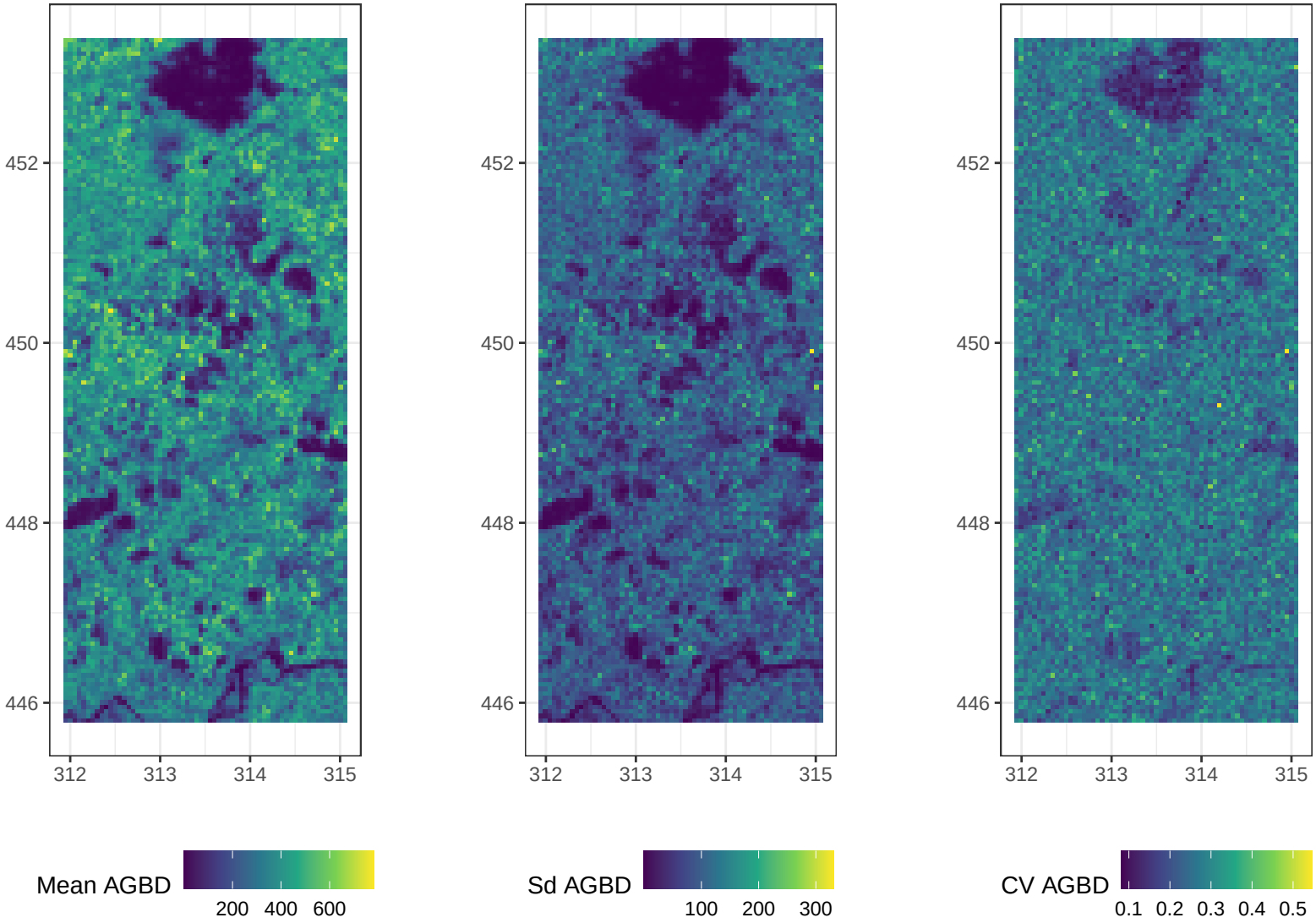
$\rightarrow$ inference results to be checked

Map predictions: arguments



- LiDAR metrics at landscape level, resolution managed inside the function
- *brms.fit* object, output of calibration step
- optional: another raster for footprint

Map predictions: output



Validation

5 What about validation ?

We suggest to validate the model by subsetting your subplot dataset in two: a calibration dataset with about 70% of the subplots, and a validation dataset with the rest. We have 4 plots divided in 16 subplots, so we need to set aside 45 subplots for the calibration step, the rest will be used to validate.

```
# let's select subplots for each step
set.seed(1234)
vector_of_subplots <- unique(dt_inf$subplot_ID)
calibration_subplots <- sample(vector_of_subplots, 45)
validation_subplots <- vector_of_subplots[!(vector_of_subplots %in% calibration_subplots)]

# then divide our complete dataset
dt_calibration <- dt_inf %>%
  filter(subplot_ID %in% calibration_subplots)
dt_validation <- dt_inf %>%
  filter(subplot_ID %in% validation_subplots) %>%
  # change in column names to make predict function work
  mutate(x = x_center) %>%
  mutate(y = y_center) %>%
  mutate(log_CHM = log(raster_metric))
```

Now we can calibrate the model (ie run the inference) with the calibration dataset:

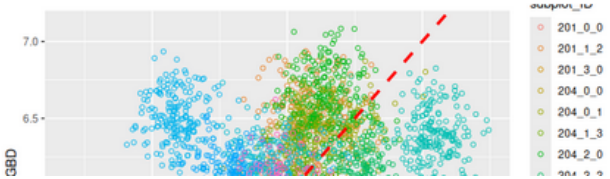
```
cal_step1 <- calibrate_model(long_AGB_simu = dt_calibration, nb_rep = 50, useCache = T,
                             plot_model = TRUE, chains = 4, thin = 20, iter = 2500,
                             warmup = 500, cores = 4)
```

Now we are going to predict the model outputs (namely AGBD) at observed data points using the parameters that we inferred at step 1 when calibrating. For that we use the predict function of brms (link to help):

```
val_step2 <- predict(cal_step1, newdata = dt_validation, ndraws = 1)
```

Now we may compare those values predicted by the model, which was calibrated with the calibration dataset, with observed values from the validation dataset:

```
dt_validation <- cbind(dt_validation, logAGBD_pred = val_step2[,1])

ggplot(dt_validation, aes( x = log(AGBD), y = logAGBD_pred, colour = subplot_ID))+
  geom_point(shape = 1)+
  geom_abline(colour = "red", linetype = 2, linewidth = 1.2)+
  xlab("Observed log AGBD")+
  ylab("Predicted log AGBD")
```



- for now, no embedded validation procedure in the package
- in the vignette we propose a example of validation method
- to be discussed this week?

Code for BIOMASS pipeline

5118 lines of R code in the entire BIOMASS package

```
[lamonica@T702 biomass]$ cloc BIOMASS
     490 text files.
     236 unique files.
     563 files ignored.

github.com/AlDanial/cloc v 1.98  T=0.57 s (415.3 files/s, 263912.0 lines/s)
-------------------------------------------------------------------------------
Language                     files          blank        comment           code
-------------------------------------------------------------------------------
CSV                             78              1              0         108210
HTML                            56           1621              0           7566
CSS                              6           3176             41           7160
JavaScript                      13           2117           1975           7110
R                               60           1590           2474           5118
YAML                             7             66             41            357
Text                             3             39              0            322
Rmd                              2            255            340            302
Markdown                         1             23              0             40
JSON                             8              0              0             28
SVG                              2              0              1             14
-------------------------------------------------------------------------------
SUM:                           236           8888           4872         136227
-------------------------------------------------------------------------------
```

Yet...

```r
## FULL BIOMASS PIPELINE
## load data
load("data/plot_data_mbalmayo.rda")

## HD model
# Fit global height-diameter model
hd_model <- BIOMASS::modelHD(D = plot_data$height_diameter$D, H = plot_data$height_diameter$H, method = "log1", useWeight = TRUE)
# Check model fit
hd_model$model

## Compute AGB with error propagation
error_prop <- AGBmonteCarlo(D = plot_data$individuals$dbh, Dpropag = "chave2004", HDmodel = hd_model, WD = plot_data$individuals$taxa_mean_wood_density, errWD = plot_data$individuals$taxa_sd_wood_density)
AGB_by_plot <- summaryByPlot(AGB_val = error_prop$AGB_simu, plot = plot_data$individuals$plot_name, drawPlot = TRUE)

## Plot coordinates projection
plot_coord <- plot_data$coordinates %>%
  select(c("typevalue_ddlat", "typevalue_ddlon", "Xrel", "Yrel", "plot_name")) %>%
  relocate(plot_name)

coord_utm <- latlong2UTM(data.frame(plot_coord$typevalue_ddlon, plot_coord$typevalue_ddlat))
plot_coord <- plot_coord %>%
  mutate(Xutm = coord_utm$X, Yutm = coord_utm$Y)

check_plot_trust_GPS <- check_plot_coord(corner_data = plot_coord, longlat = c("typevalue_ddlon", "typevalue_ddlat"), rel_coord = c("Xrel", "Yrel"), trust_GPS_corners = F, plot_ID = "plot_name", draw_plot = F)

multiple_checks <- check_plot_coord(corner_data = plot_coord, longlat = c("typevalue_ddlon", "typevalue_ddlat"), rel_coord = c("Xrel", "Yrel"), trust_GPS_corners = F, plot_ID = "plot_name", draw_plot = F, prop_tree = "dbh",
  tree_data = plot_data$individuals, tree_plot_ID = "plot_name", tree_coords = c("x_rel_corr","y_rel_corr"))

## Divide plots to get 0.25 ha surface plots
multiple_subplots <- divide_plot(grid_size = 50, corner_data = corner_data, rel_coord = c("x_rel", "y_rel"), proj_coord = c("x_proj","y_proj"), corner_plot_ID = "plot_ID", tree_data = plot_data$individuals,
                                 tree_plot_ID = "plot_name", tree_coords = c("x_rel_corr","y_rel_corr"))

## AGB estimates and their uncertainty per subplot
subplot_AGBD <- subplot_summary(subplots = multiple_subplots, AGB_simu = error_prop$AGB_simu)

## load CHM
chm_mba_small <- rast("/home/lamonica/Documents/biomass/data_studysitestests_geostats/cafriplots/CHM_Aout_2024_mba.tif")

## 0.25 ha median CHM
raster_summary <- subplot_summary(subplots = multiple_subplots, ref_raster = chm_mba_small, raster_fun = median, na.rm = T)

## gather data for agbd-chm model calibration
chm_subplot <- raster_summary$tree_summary %>%
  rename(raster_metric = CHM_Aout_2024_mba_median)
agbd_subplot <- subplot_AGBD$long_AGB_simu
df_inference <- agbd_subplot %>%
  left_join(chm_subplot, by = "subplot_ID") %>%
  arrange(subplot_ID)

## calibrate model
test_cal <- calibrate_model(long_AGB_simu = df_inference, nb_rep = 20, useCache = FALSE, plot_model = TRUE, chains = 4, thin = 20, iter = 2500, warmup = 500, cores = 4)
summary(test_cal)

## predict with chm set to 0.25 ha resolution
library(doFuture)
plan(multisession, workers = 4)
test_pred <- predict_map(fit_brms = test_cal, pred_raster = chm_mba_small, grid_size = 50, raster_fun = median, n_cores = 4, n_post_draws = 20, alignment_raster = NULL)

60:1  (Top Level) ÷                                                                                                                    R Script
```

... the full pipeline can be less than 60 lines, comments included!

Future work, this week and later on

- More robust quality check with single or multiple dates
- Implementation of one or more new approaches for estimating height in the absence of field data
- Taking into account positioning uncertainty on individual trees
- Management of circular plots in the spatialisation pipeline for estimates
- Flexibility (functional form, predictive metric) of the AGBD-LiDAR model (implemented in v3.0)
- Temporal BIOMASS (multiple census and LiDAR acquisitions)

Authors (v2-v3) and funders

Dominique Lamonica, Arthur Bailly, Jérome Chave, Guillaume Cornu, John Godlee, Fabian Fischer, Arthur Péré & Maxime Réjou-Méchain

European Spatial Agency

Research Institute for Development

the French agricultural research and cooperation organization